

Towards Trustworthy Augmented Reality in the Metaverse Era: Probing Manipulative Designs in Virtual-Physical Commercial Platforms

Esmée de Haas
KAIST

Huang Yiming*
Hong Kong University of
Science and Technology
Pan Hui
Hong Kong University of
Science and Technology
(Guangzhou)

Carlos Bermejo
Hong Kong University of
Science and Technology
Lik-Hang Lee†
The Hong Kong Polytechnic
University, and KAIST

Zijun Lin
London School of Economics
and Political Science

ABSTRACT

E-commerce has become an important activity where new advances in technology shape the shopper experience. At the same time, the metaverse is seen as the next milestone to revolutionize the e-commerce experience, where immersion, realism, and ubiquity are its main features. However, under such circumstances, manipulative designs to ‘trick’ users toward intended choices or outcomes can become more effective. This paper sheds light on the design space of manipulative techniques in e-commerce applications for the metaverse, reinforcing our understanding of interface design guidelines and counteracting malicious practices.

Index Terms: Human-centered computing—Human computer interaction—Interaction paradigms—Mixed/ augmented reality & Virtual reality; Human-centered computing—User experience—User experience & usability—

1 INTRODUCTION

Metaverse is a digital ecosystem [2] that involves immersive technologies, including Virtual Reality (VR), Augmented Reality (AR), and the variations between AR and VR. New technological advances (e.g., machine learning) allow people to have a highly customised avatar [3] to live in the metaverse with the voice they prefer [5].

E-commerce has been growing steadily in recent years, and its adoption is tightly linked with advances in technology. E-commerce presents several advantages to traditional retail stores. However, the increased immersivity and interactivity of the metaverse could be used to create more sophisticated manipulative designs (MDs). The levels of misleading and surreptitiousness in MDs can be exaggerated by AR [8]. These manipulative designs, also known as dark patterns (DPs) [7], have been broadly reported in the research community [1, 4], and we can usually depict them in online advertisements, mobile applications, and e-commerce [4]. Despite the challenges in perceiving and regulating these manipulative/deceptive designs, we observe that regulators, such as the California Consumer Privacy Act (CCPA), are taking an active role in protecting users against such malpractices. We can foresee that these manipulative designs in the metaverse will bring more challenges to protecting the users against them.

2 MANIPULATION OF EXISTING APPS

Augmented views, e.g., augmented reality (AR), augmented virtuality (AV), allow shoppers to create a more dynamic environment surrounding the customers. We could imagine a physical store that

*Y. Huang is an intern at KAIST during the study

†corresponding author: see www.lhlee.com, email: lhleeac@connect.ust.hk

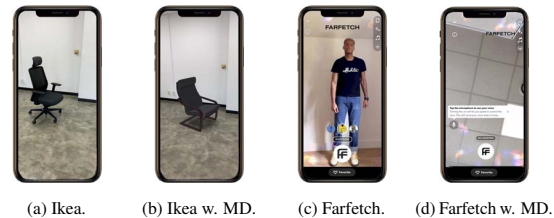


Figure 1: Examples of collected MAR app once enabling the manipulative design (MD). 1b modifies the floor surface (i.e., interface interference); & 1d enforces turning the microphone on for accessing the virtual try-on (i.e., obfuscation).

displays discounts and descriptions of targeted products in the user’s field of view of their smart glasses. Another possibility is the use of location-based triggered interactions (proxemic interactions [4]) when the users move around virtual or physical retail. These manipulations of the environment will influence the users’ purchase decisions [4, 8].

We aim to probe the current manipulative designs (if any) that brands and companies implement in their mobile augmented reality (MAR) apps. Then, we depict the data collection procedure and our findings regarding commercially available MAR apps. We focus on apps with the following features: (i) AR experience, (ii) shopping features (users can buy the displayed items), and (iii) trending online websites due to their AR features. Some findings include any change in visuals that differ from the actual camera perspective on the user’s device (Figure 1a & b). We also find that some apps inquire the user to do any action to use the AR feature (Figure 1c & d). Inquiries included were either for using the microphone, enabling notifications, or looking into the camera to do some action. One of the findings was that the app required the user to smile to the camera to continue the application usages.

3 USER STUDY

Although the above results reveal the current techniques of MD in MAR apps, we need further exploration, as the recorded approaches do not take all the advantages of AR environments, such as immersion, realism, and mobility. These features of AR differ from other traditional scenarios, such as computers and smartphones, and thus open the possibilities for a new design space for MDs. Also, the current AR headsets own limited Field-of-View and hence deliver ‘weak’ visuals that are relatively less prominent than auditory cues. Nonetheless, we know a little regarding the effect of auditory cues related to MD in MAR apps. Thus, we conducted a study to evaluate the auditory effects of MDs, specifically in MAR apps.

Study design. We implemented two fully working MAR apps, where the participants can select between two items they would like to buy (Figure 2). In the first app (flowerpot), we display two flower-

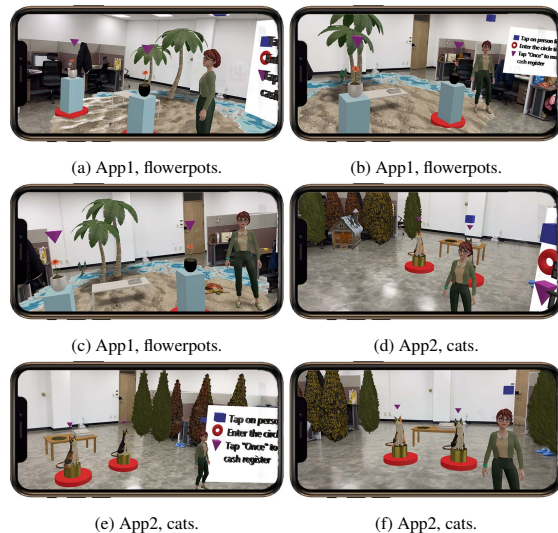


Figure 2: MAR scenarios: (a–c) App 1: flowerpots; and (d–f) cats (Demo video in [6]).

pots with different colours. In the second app (cat), the participants select one of the two displayed cats (also with different colours). We design a factorial study that compares the effects of MDs on the participants' decision-making: 2 scenarios (between-participants: two MAR apps) $\times$ 2 sides (between-participants: unpleasant sound on the right or left object) $\times$ 2 conditions (within-participants: manipulative design (MD), no manipulative design (NoMD)).

Experimental conditions. We include two audio sounds for each flowerpot for the first app: (i) screeching bird sounds and (ii) calming (sea) wave sounds. We assigned the MD to any of the flowerpots for each participant. Both flowerpots have no sound in conditions with no MDs (NoMD). In the second app, we follow a similar technique as above, where two different cats own the following audio feedback: (i) the hissing of a cat and (ii) the meowing of a cat. For both apps, the sound is triggered when the participants are in the item's proximity (cat or flowerpot). As our main focus was the audio effects on the participants' responses, we randomise the visual designs of flowerpots and cats accordingly.

Participants. We recruited a total of 40 participants: 9 females, 30 males, and one participant who preferred not to disclose this information. The participants' age ranges from 18 to 44 years old. Amongst the participants, 87.5% were 18–24 years old, 10% were 25–34 years old, and 2.5% were 35–44 years old. Their educational backgrounds are distributed by 72.5%, 17.5%, and 10% for bachelor's, master's, and doctoral levels, respectively. The participants rated their previous experience with MAR apps as follows: 15% excellent, 47.5% good, 27.5% average, and 10% poor. The participants rated their trust in MAR apps as 3.48 (SD: 0.84) out of 5 (max).

Apparatus. We employ an iPhone 11 Pro to run the two MAR apps (through-the-screen AR apps). We design the apps using *Adobe Aero* that generates a neat but interactive environment. We requested the participants to wear a pair of headphones (generic brand) to receive audio from the apps. We conducted our experiment in a quiet office room inside our university campus.

Task and procedures. We invited the participants to our lab in a consecutive sampling, where we first asked them permission with a consent form that included the data collection and processing policies during our experiment. After signing the consent form, we requested the participants to stand in our lab's marked (on the floor)

position. Then, we ask the participants to open the AR apps on iPhone 11 Pro and headphones. The participants were instructed to interact with the AR environment, where they could freely move. A female instructor explains how they can interact with the flowerpots & cats on the screen and encourage them to approach the items before selecting them. Next, we asked the participants to select the flowerpot & cat according to their preferences by walking near the desired object and touching the flowerpot & cat they would choose on the smartphone's screen.

4 RESULTS

We first analyse the impact of MDs on the participants' responses and evaluate the participants' perception skills to detect malicious MDs in the MAR apps they use to select an object (between two). The majority of the participants that selected the pleasant audio feedback in the MD scenarios and noticed a malicious design detected the audio feedback as the MD technique. 59% (n=16) of the participants that selected the pleasant sound objects (67.5%, n=27) detected the audio feedback MD. Overall, 72.5% (n=29) of the participants detected the MD as audio feedback provided in MD scenarios. In the NoMD scenarios, 27.5% (n=11) of the participants noticed MDs such as objects in the background (e.g., trees, table), the distance between the objects (flowerpots & cats), and the scale of the objects. Most of the participants referred to the sea wave sound as a "likeable" or "enjoyable" sound, while the screeching bird sound was often described as "scary", "loud" or "uncomfortable". Some participants even mentioned not noticing the difference in the colour of each flowerpot, because of their distraction towards the sounds.

5 CONCLUSION

It is vital to emphasise the potential risks in MAR and proposes several guidelines for the future design of MAR while also proposing methods to protect the user against the possible threats that can occur in the MAR environment. The metaverse will have a significant impact on users' shopping experience. However, all these advances and experiences in e-commerce will bring several challenges regarding manipulative designs and how to protect shoppers against them. Thus, researchers, regulatory organisations, and developers should be fully involved to avoid such practices.

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